Supplementary material for
Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools

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1 Computational evidence design

The computational study separates formulation validation, nominal method comparison, mechanism-specific controls, manufacturing-factor analysis, parameter sensitivity, and distribution-shift evaluation. Table 1 gives the complete prespecified design. A run is identified by the instance, method, solver seed, training seed where applicable, and experimental profile. All observations are retained, including time-limited runs.

Table 1: Prespecified computational suites.

Suite	Design	Runs
Validation	Four small physical cases, one exact solve each	4
Nominal	20 instances; two solver baselines with five solver seeds and five learned configurations with five solver and five training seeds	2700
Two-stage stability	20 instances, three refinement profiles, five solver seeds, and five training seeds	1500
Factorial DOE	$4 \times 5 \times 3$ wafer-count, recipe-composition, and processing-scale cells with five solver seeds	300
SPBS	48 instances, automatic versus no warm start, ten matched solver seeds	960
Sensitivity	Eight one-factor settings with ten solver seeds	80
Distribution shift	Nine out-of-distribution instances; two solver baselines and five learned configurations with matched solver and training realizations	729
Total	Excluding the four validation cases	6269

All 6269 formal benchmark runs and four validation runs completed without execution failure. Every saved incumbent used in the reported statistics was checked independently against the corresponding mathematical instance. The distribution-shift suite contains 729 retained time-limited runs; every run returns an independently valid incumbent.

2 Indexed MILP specification

This section gives the complete constraint architecture used to instantiate the finite-horizon model. Let $\mathcal{W}$, $\mathcal{P}$, $\mathcal{C}$, $\mathcal{K}$, $\mathcal{O}$, and $\mathcal{R}$ denote product wafers, product groups, chambers, candidate recipe positions, operations, and unit-capacity resources. $\mathcal{K}(p)$ contains positions compatible with group p; $\mathcal{A}$ is the set of technological arcs; and $\mathcal{O}_r$ contains operations that may use resource r. For chamber c, $\mathcal{J}_c^{\mathrm{DW}}$ contains dummy-wafer demands and $\mathcal{T}_c$ its reusable physical dummy wafers. Cleaning epochs are $\mathcal{E}_c$.

The binary variables are assignment x_{pk}, position activation u_k, operation activation b_o, cycle-to-epoch assignment a_{vce}, epoch use z_{ce}, dummy identity $g_{j\tau}$, and resource order $y^r_{oo'}$. Continuous variables s_o, e_o are event start and completion times. The fixed-duration, activation, and assignment relations are

$$\sum_{k\in\mathcal{K}(p)} x_{pk} = 1 \qquad (p \in \mathcal{P}), \tag{1}$$

$$x_{pk} \le u_k, \quad u_{k+1} \le u_k, \quad b_o = \chi_o(x, u, a) \qquad (o \in \mathcal{O}), \tag{2}$$

$$0 \le s_o \le M b_o, \quad e_o = s_o + d_o b_o \qquad (o \in \mathcal{O}), \tag{3}$$

where the binary affine incidence map χ_o activates exactly the events belonging to the selected 4×1 side, 2×2 chain position, or cleaning cycle. The full route logic is generated from the following active event chains:

Structure	Ordered active events
Product route	load port → ATR → aligner → upper load lock → VTR load → chamber process → VTR unload → lower load lock → ATR → load port
4×1	front load → rotate/back load → two-side process → back unload → rotate/front unload
2×2	dummy/product head → adjacent product bridges → product/dummy tail; consecutive positions share the prescribed product pair and may split only at a cleaning boundary
Cleaning	front dummy load → back dummy load → clean process → back unload → front unload

For every active technological arc and every unordered pair competing for a resource, timing feasibility is imposed by

$$s_{o'} \ge e_o + \delta_{oo'} - M(2 - b_o - b_{o'}) \qquad (o, o') \in \mathcal{A}, \tag{4}$$

$$s_{o'} \ge e_o + \rho^r_{oo'} - M(1 - y^r_{oo'}) - M(2 - b_o - b_{o'}), \tag{5}$$

$$s_o \ge e_{o'} + \rho^r_{o'o} - M y^r_{oo'} - M(2 - b_o - b_{o'}) \qquad \{o, o'\} \subseteq \mathcal{O}_r. \tag{6}$$

The setup $\rho^r_{oo'}$ includes the empty reposition time from the endpoint of o to the origin of o'. These disjunctions cover the atmospheric and vacuum robots, aligner, individual load-lock slots, rotary chambers, and physical dummy-wafer identities.

Dummy identity and cleaning are linked by

$$\sum_{\tau\in\mathcal{T}_c} g_{j\tau} = b_j \qquad (j \in \mathcal{J}_c^{\mathrm{DW}}), \tag{7}$$

$$\sum_{e\in\mathcal{E}_c} a_{vce} = b_v, \quad a_{vce} \le z_{ce}, \quad \sum_{v} a_{vce} \le H z_{ce}, \tag{8}$$

$$z_{c,e+1} \le z_{ce}, \qquad \sum_{e} e\, a_{vce} \le \sum_{e} e\, a_{v+1,c,e}. \tag{9}$$

If $g_{j\tau} = g_{j'\tau} = 1$, the same resource disjunction is activated for dummy uses j and j', preventing overlapping reuse. Epoch transitions activate exactly one intervening dummy-only cleaning route. The last inequality makes epoch indices nondecreasing along chamber positions; the adjacency equations in χ_o permit a 2×2 split only at that transition.

Residence and completion constraints are

$$0 \le s_{\text{out}(w,r)} - e_{\text{in}(w,r)} \le \overline{R}_{wr}, \tag{10}$$

$$C_{\max} \ge e_w^{\text{return}} \qquad (w \in \mathcal{W}). \tag{11}$$

Stage one minimizes $C_{\max}$. Stage two fixes all retained integer values, adds $C_{\max} \le \widehat{C} + 10^{-6}$, and minimizes

$$\Phi = L_{\text{sum}} + 10L_{\max} + 25D_{\text{cadence}}, \tag{12}$$

where L_{sum} and $L_{\max}$ aggregate the model-native wait/idle variables kept tight by the linear objective and

$$D_{\text{cadence}} = \sum_{m,c,e,\ell \ge 2} \left| (h_{mce,\ell+1} - h_{mce,\ell}) - (h_{mce,\ell} - h_{mce,\ell-1}) \right|. \tag{13}$$

A soft-cap excess term is supported by the formulation but has weight zero in the reported profile. Hence the second stage preserves the stage-one makespan within numerical tolerance; if stage one is time-limited, this statement applies to its retained incumbent rather than to an unproved global optimum.

The same inequalities exclude scheduling deadlock in each generated complete schedule. Any circular wait would be a directed cycle in the combined technological and selected resource-order graph. Summing its positive separation inequalities yields a contradiction, while terminal constraints require all material routes to return. Thus every feasible MILP solution is a deadlock-free offline trajectory under the modeled operating conditions.

3 Independent validation and numerical tolerance

The validator reloads each saved solution, fixes the primary decision values, and checks the active physical model independently of the learning policy. The checks cover routing and stage order, chamber and slot exclusivity, robot and load-lock use, recipe grouping, cleaning, residence-time limits, and vacuum-zone dummy-wafer capacity. Continuous residuals use the solver tolerance 10^{-6} plus a 10^{-12} serialization allowance, i.e., an effective threshold of 1.000001×10^{-6}. Binary and integer requirements are not relaxed. This allowance only accounts for the decimal text round trip of an otherwise accepted solver solution.

4 Event-time reconstruction of schedule stability

Schedule-stability metrics are reconstructed from physical event times rather than auxiliary objective variables. This makes the definition identical under all three refinement profiles. For product wafer w, define six nonnegative route waits:

$$q_{w,1} = [s_w^{\text{PM}} - e_w^{\text{VTRload}}]_+, \qquad q_{w,2} = [s_w^{\text{VTRunload}} - e_w^{\text{PM}}]_+, \tag{14}$$

$$q_{w,3} = [s_w^{\text{AL}} - e_w^{\text{ATR,LP}\to\text{AL}}]_+, \qquad q_{w,4} = [s_w^{\text{ATR,AL}\to\text{LL}_\text{u}} - e_w^{\text{AL}}]_+, \tag{15}$$

$$q_{w,5} = [s_w^{\text{VTRload}} - e_w^{\text{LL}_\text{u}}]_+, \qquad q_{w,6} = [s_w^{\text{ATR,LL}_\text{l}\to\text{LP}} - e_w^{\text{LL}_\text{l}}]_+, \tag{16}$$

where $[x]_+ = \max(0, x)$ and s and e denote stage start and end times. The route-total and route-maximum metrics are

$$Q_{\text{total}} = \sum_{w} \sum_{k=1}^{6} q_{w,k}, \qquad Q_{\max} = \max_{w,k} q_{w,k}. \tag{17}$$

Processing cadence is reconstructed from the ordered start times of active full-flow and mixed-flow chamber batches. For adjacent-start intervals $d_1, \ldots, d_m$, cadence variation is $\mathrm{CV}(d) = \mathrm{sd}(d)/\,\mathrm{mean}(d)$. The cadence-deviation sum is the sum of the absolute deviations used by the linear refinement objective after the discrete batch structure has been fixed.

Table 2: Instance-first means in the controlled two-stage experiment.

Profile	Route-total	Route-maximum	Cadence CV	Cadence deviation
Full	2596.01	296.33	**0.130**	**22.04**
Quadratic objective (linear off)	**2469.47**	**293.27**	0.134	60.18
Stage2-off	5398.60	952.57	0.216	370.41

Relative to stage2-off, the full profile reduces route-total wait by 51.91%, route-maximum wait by 68.89%, cadence CV by 39.70%, and cadence deviation by 94.05%; all four comparisons remain significant after Holm adjustment. The linear objective also reduces cadence deviation by 63.38% relative to the quadratic profile. Its route-total, route-maximum, and cadence-CV differences from the quadratic profile are not Holm-adjusted significant.

5 Statistical protocol

The manufacturing instance is the unit of inference. Repeated solver and training realizations are first averaged within each instance and method or profile. Mean differences are reported with a nonparametric 95% bootstrap interval. Paired two-sided Wilcoxon signed-rank tests use the same instance pairs, and all planned contrasts within an analysis family are adjusted by the Holm procedure. Failure, incumbent, and optimality indicators are retained as binary outcomes. Primal–dual integral (PDI) is the primary anytime measure; lower values are better. Time-limited runs remain in PDI, time, incumbent, and failure summaries.

Specifically, if $G(t)$ is SCIP's normalized primal–dual gap at time t,

$$\mathrm{PDI}(T) = \int_0^T G(t)\, dt. \tag{18}$$

Terminal gap is $G(T)$. Reported solution time covers model loading, the primary SCIP solve, the cut-selection callback, conditional second-stage refinement, and solution writing inside the common measured run; offline SPBS generation is excluded and is evaluated separately by the paired SPBS suite.

6 Solver, learning, and decoding configuration

All nominal methods receive the same precomputed, independently verified SPBS incumbent. Consequently, the 2700-run comparison isolates cut-selection and decoding policies under common primal guidance; it does not estimate an SPBS–policy interaction. The seven configurations are:

SCIP default Native SCIP cut selection with the shared SPBS incumbent.

ACS A four-score SCIP selector with direction, efficacy, integer-support, and objective-parallelism weights $(0.25, 0, 0.50, 0.25)$.

HEM-13D The adapted 13-feature hierarchical policy with greedy decoding.

HEM-13D+Beam The identical trained HEM policy parameters with width-three beam decoding.

23D Features Only The reconstructed 23-feature policy with greedy decoding and no role-based reranking.

Structure+Greedy The trained proposed policy parameters and role-based completion with greedy decoding.

RDMCT-HBS The same trained proposed policy parameters and completion rule with width-three beam decoding.

Each candidate cut has 13 generic attributes: objective parallelism, efficacy, support, integer support, normalized violation, and the mean, maximum, minimum, and standard deviation of cut and objective coefficients. Ten added descriptors are the coefficient-mass shares on binary, general-integer, objective-active continuous, auxiliary-continuous, and other variables, followed by bounded-variable share, positive-coefficient share, discrete–continuous coupling, dominant-role share, and normalized role entropy. Candidates are first screened deterministically by efficacy to at most 128 rows, and at most 16 are selected.

The 23-dimensional input passes through layer normalization, separate projections of the generic and role blocks, gated fusion, a one-layer LSTM, and pointer attention. A high-level policy chooses the selection fraction; a low-level pointer policy with an end token generates an ordered, nonrepeating sequence. Beam width is three, with cumulative joint log probability and no length normalization. Structural completion retains at most $\lceil 0.25k \rceil$ policy anchors for target cardinality k and greedily fills from a pool of at most $2k$ candidates. Its quality, policy, role coverage, and representativeness weights are 0.35, 0.20, 0.25, and 0.20. The associated $(1 - 1/e)$ property applies only to the residual monotone-submodular completion problem with fixed anchors and pool, not to global cut selection.

Training uses five training instances and five independent training seeds. For each seed, two workers collect eight trajectories per epoch for 60 epochs; each trajectory has a 30-s/1024-MB SCIP budget, including a 5-s conditional refinement. A central Adam update uses an exponential-moving-average baseline with coefficient 0.9. Low- and high-level learning rates are 10^{-4} and 5×10^{-4}; the gradient norm is capped at 2, and the learning rate is multiplied by 0.96 every ten epochs. The return is negative SCIP PDI. This is an A3C-inspired parallel hierarchical REINFORCE backbone with no learned critic and no asynchronous parameter update.

7 SPBS construction and acceptance

SPBS fixes deterministic assignment, activation, cleaning-epoch, load-lock slot, and selected ATR/aligner/load-lock/VTR order roles in a temporary zero-objective feasibility MIP. If no solution is found, a fixed sequence of profiles progressively releases the most globally sensitive resource orders and then slot choices. Each attempt stops at its first feasible solution. The selected physical schedule is reconstructed in the full model with secondary variables free and is accepted only after a complete SCIP feasibility check. The production MILP receives only this incumbent; no temporary fixing survives. All formal starts are additionally keyed to the exact LP hash.

8 Hardware, software, and budgets

Experiments ran under Windows 11 Pro on an Intel Core Ultra 7 265KF (20 cores), 31.7 GB RAM, and an NVIDIA GeForce RTX 5060 Ti with 8 GB memory. The evaluation environment uses Python 3.12.12, SCIP 10.0.1 through PySCIPOpt 6.1.0, and PyTorch 2.11.0+cu130. The code does not impose a solver thread count, so no single-thread assumption is used in the analysis.

Table 3: Formal per-run solver budgets.

Suite	Time (s)	Memory (MB)	Stage-two allocation
Validation	600	2048	up to 60 s
Nominal	600	2048	540 s primary + up to 60 s
Two-stage stability	600	2048	profile-specific; see text
Factorial DOE	600	2048	540 s primary + up to 60 s
SPBS	600	2048	540 s primary + up to 60 s
Sensitivity	600	2048	540 s primary + up to 60 s
Distribution shift	1200	4096	1140 s primary + up to 60 s

In the two-stage control, Full and Quadratic-objective (linear-off) reserve 60 s for linear and quadratic refinement, respectively; Stage2-off assigns all 600 s to the primary solve. Therefore PDI and time from that control have unequal primary horizons and are not used as causal evidence for the second stage. The reported route-wait and cadence conclusions are reconstructed from the saved event schedules under identical definitions.

9 Independent SPBS warm-start experiment

Table 4 reports the controlled SPBS comparison. Automatic SPBS provides a valid incumbent in every run and reduces PDI by 7194.27 (26.19%). The matched instance-level effects on incumbent availability and PDI both remain significant after multiplicity adjustment.

Table 4: SPBS warm-start comparison over 48 instances and ten matched seeds.

Condition	Runs	Incumbent (%)	Optimal (%)	Mean PDI
Automatic SPBS	480	**100.00**	54.58	**20278.78**
No warm start	480	51.67	51.67	27473.05

The incumbent-rate difference is 48.33 percentage points (95% CI $[34.58, 62.29]$; Holm $p = 2.55 \times 10^{-5}$). The PDI difference is 7194.27 (95% CI $[5145.78, 9317.61]$; Holm $p = 3.46 \times 10^{-8}$). The optimal-rate and elapsed-time differences do not remain significant after Holm adjustment.

10 One-factor parameter sensitivity

Table 5 reports all eight prespecified settings. Each row contains ten default-SCIP runs and every run provides an independently valid incumbent. The experiment evaluates operational solvability under parameter perturbations; because it contains one method, it is not used for a cross- method robustness ranking.

Table 5: One-factor sensitivity results for the 24-wafer 1:1 mixed-flow case.

Setting	Optimal (%)	Mean PDI	Mean time (s)
Nominal	20	40728.91	515.09
Motion scale 0.8	40	37126.93	494.55
Motion scale 1.2	0	42478.32	539.79
Processing scale 0.8	60	33961.82	459.63
Processing scale 1.2	10	38663.67	508.40
Cleaning interval 3	10	38429.95	517.26
Cleaning interval 10	40	28899.41	497.91
Vacuum-zone dummy-wafer inventory 10	20	38152.52	484.80

11 Distribution-shift stress test

The OOD suite contains nine larger 40–64-wafer configurations. Its complete 729-run matrix has no duplicate key, execution error, solution-write error, or invalid saved solution. Table 6 reports the instance-first summaries. Every method attains 100% incumbent availability and 0% proven optimality under the common horizon.

Table 6: Complete OOD comparison over nine larger configurations.

Method	Runs	Mean PDI	Mean gap (%)
SCIP	27	73670.99	171.35
Adaptive cut selection	27	**70915.54**	**160.78**
HEM-13D	135	73003.59	165.88
HEM-13D+Beam	135	73165.58	167.04
23D Features Only	135	74187.22	169.96
Structure+Greedy	135	73507.76	168.54
RDMCT-HBS	135	73919.15	170.02

Adaptive cut selection has the lowest OOD mean PDI. RDMCT-HBS has a mean terminal gap 0.78% below SCIP. None of the OOD pairwise contrasts remains significant after global Holm adjustment. The complete suite establishes the scalable feasibility of the shared SPBS–MILP–SCIP pipeline and complements the observed nominal-range mean anytime advantage reported in the main paper.

12 Factorial-model diagnostics

The factorial design contains 60 complete cells and five solver seeds per cell. The prespecified PDI model has $R^2 = 0.9980$. Wafer count and recipe composition exhibit strong interactions, whereas all prespecified processing- scale main-effect and interaction confidence intervals include zero. Figure 1 provides the residual and influence diagnostics.

13 Reproducibility record

The reproducibility package preserves the instance manifest, solver and training seeds, method configuration, time and memory limits, trained-model metadata, raw row records, saved incumbent solutions,

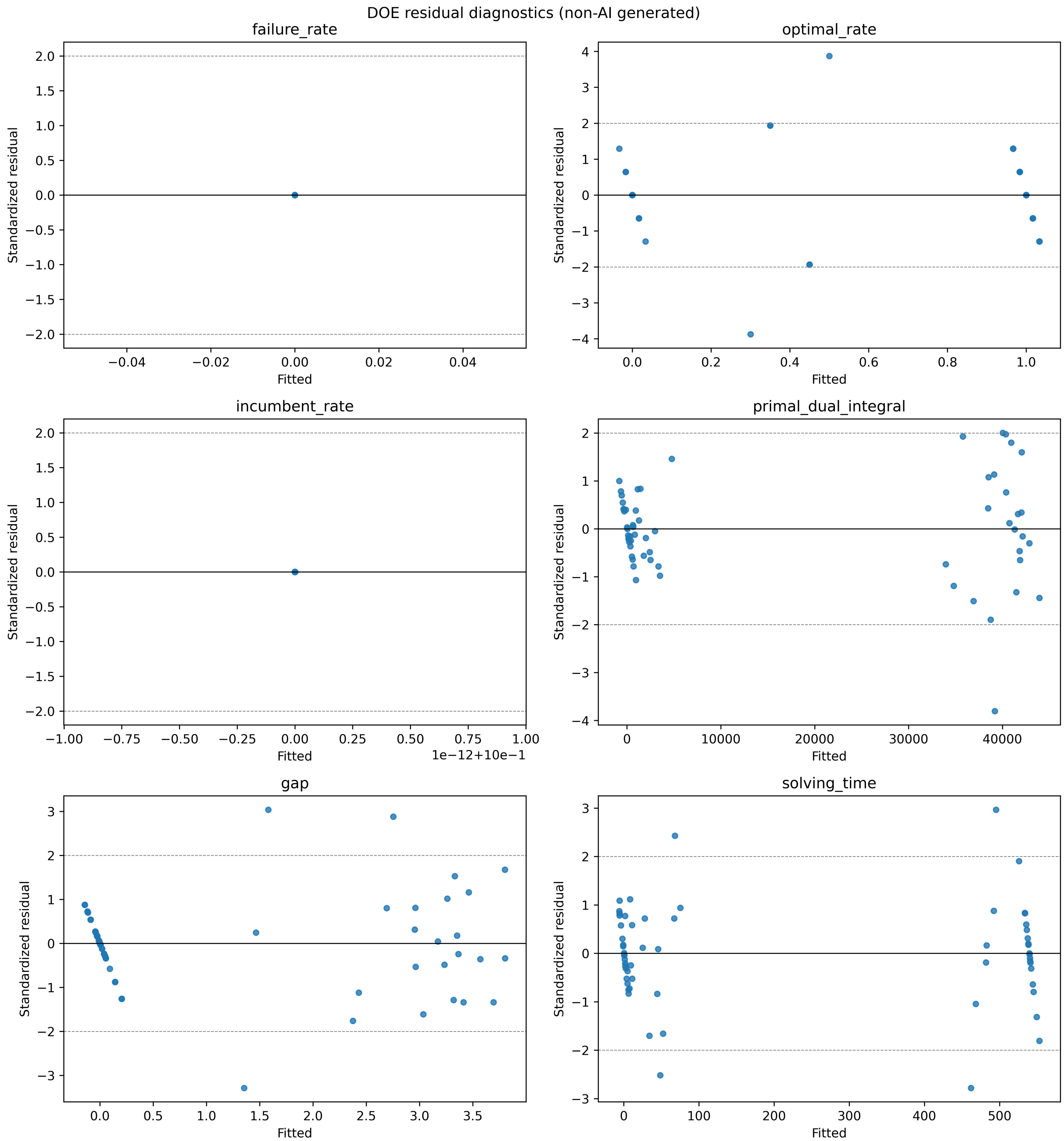

Figure 1: Diagnostic plots for the prespecified factorial PDI model fitted to the complete 300-run design.

independently derived manufacturing metrics, and machine-readable statistical summaries. Every formal suite is accepted only when its expected matrix is complete, duplicate keys and execution errors are absent, saved solutions are covered one-to-one by independent validation, and the statistical input count equals the frozen raw count.